ApexStrykerz Parts Inventory (September 2026)

Compiled from four orders: the REV Robotics order, the DigiKey order, and the mechanical team's goBILDA and AndyMark order. Kit contents come from each vendor's published bill of materials. Legal limits from the BIOBUZZ manual: **8 motors, 8 servos, 1 battery, 1 Control Hub (+ up to 1 Expansion Hub)** on the robot.

Assumption to confirm: the goBILDA Strafer kit appears on two lists. This inventory counts **one** kit, the one already built. If a second unopened kit exists, add 4 more 312 RPM motors and 4 mecanum wheels.

1. Control system and power

Item	SKU	Qty	Source	Notes
REV Control Hub	REV-31-1595	2	REV order, DigiKey	Only one may be on the robot. Second is a spare. Built-in IMU.
REV Expansion Hub	REV-31-1153	1	DigiKey	Adds 4 motor + 6 servo ports.
REV Driver Hub	REV-31-1596	2	REV order, DigiKey	Driver Station. One spare.
Driver Hub Battery	REV-31-1876	2	REV order, DigiKey	
REV USB PS4-compatible Gamepad		2	REV order	
12V Slim Battery	REV-31-1302	3	2 REV order, 1 in Starter Kit	Legal main battery.
AndyMark FTC Flat Pack Battery	am-5290	1	Robits kit	Legal main battery. 4 batteries total.
REV Battery Charger	REV-31-1299	2	REV order, Starter Kit	
AndyMark FTC Battery Charger	am-5473	1	Robits kit	
REV Switch Cable and Bracket	REV-31-1387	3	REV order, DigiKey, Cable Bundle	Legal main power switch.
AndyMark FTC Power Switch w/ bracket	am-4969	1	Robits kit	Legal main power switch.

Item	SKU	Qty	Source	Notes
SRS Programmer	REV-31-1108	2	REV order, Starter Kit	Programs Smart Robot Servos.
Logitech C920 webcam	C920	1	DigiKey order	UVC, legal for AprilTag vision.

2. Motors (12 total, 8 allowed on the robot)

Motor	SKU	Qty	Output	Where it came from	Good for
goBILDA 5203 Yellow Jacket 19.2:1, 312 RPM, 8mm REX	5203-2402-0019	4	312 RPM, 24.3 kg·cm	Strafer kit	Drivetrain (installed)
goBILDA 5203 Yellow Jacket 50.9:1, 117 RPM, 8mm REX	5203-2402-0051	1	117 RPM, 62 kg·cm	DigiKey	Spare; v2 lift (needs goBILDA cables)
REV HD Hex Motor + UltraPlanetary gearbox kit	REV-41-1600	3	6000 RPM bare, gearbox stages 3:1, 4:1, 5:1 mix and match	Starter Kit	Flywheel (bare or 3:1), lift
REV Core Hex Motor	REV-41-1300	1	125 RPM, 3.2 N·m	Starter Kit	Intake flapper (5 mm hex output)
AndyMark NeveRest Hex 19.2	am-5442	2	340 RPM	Robits kit	Intake, feeder
AndyMark NeveRest Hex 50.9	am-5443	1	128 RPM	Robits kit	Slow mechanisms

All are on the legal motor list (R501). Mixed shaft standards: goBILDA is 8mm REX, REV is 5mm hex, AndyMark NeveRest Hex is 5mm hex (the Robits system then adapts to 3/8 in hex).

3. Servos (10 total, 8 allowed)

Servo	SKU	Qty	Source	Notes
REV Smart Robot Servo V2 (Balanced)	REV-41-3334	6	3 Starter Kit, 2 REV order, 1 DigiKey	Programmable, continuous or angular. Legal.

Servo	SKU	Qty	Source	Notes
AndyMark Programmable Torque Servo	am-4954a	2	Robits kit	Legal (listed as example in R502).
AndyMark Programmable Speed Servo	am-4955a	2	Robits kit	Confirm ≤ 8 W / ≤ 4 A at inspection.
Servo accessories			Starter Kit	2 Aluminum Servo Horns V2, 2 Aluminum Double Servo Arms, 4 Servo Gear Adapters, 4 Servo Shaft Adapters

4. Sensors (FTC Sensor Bundle REV-45-1885, plus hubs)

Sensor	SKU	Qty	Use
2m Distance Sensor	REV-31-1505	2	Ball detection in hopper, wall distance
Color Sensor V3	REV-31-1557	2	Pollen vs Nectar, our color vs theirs
Touch Sensor	REV-31-1425	2	Limit switches
Magnetic Limit Switch	REV-31-1462	2	Limit switches
Digital LED Indicator (red/green)	REV-31-2010	4	Driver feedback
IMU	built into Control Hub	1	Heading
Webcam	Logitech C920	1	AprilTags on the Hive cells

5. Cables and wiring (FTC Cable Bundle REV-45-1901 plus kits)

Item	SKU	Qty
36 in PWM cable	REV-11-1130	12 (2 packs bundle + 1 pack kit)
24 in PWM cable	am-0694a	2
36 in PWM cable	am-0695b	2
PWM cable clips	REV-11-1229	10
Resistive grounding strap	REV-31-1269	2

Item	SKU	Qty
Sensor splitter cable	REV-31-1386	2
JST PH 4-pin joiner board	REV-31-1388	12 (8 bundle + 4 sensor bundle)
XT30 extension 30 cm	REV-31-1392	2
XT30 extension 50 cm	REV-31-1394	4
JST PH 4-pin sensor cable 50 cm	REV-31-1408	8
JST PH 4-pin sensor cable 100 cm	REV-31-1409	8 (4 bundle + 4 sensor bundle)
JST VH 2-pin motor cable	REV-31-1413	8
JST VH 2-pin motor cable 100 cm	REV-31-1526	4
JST VH 2-pin joiner board	REV-31-1429	8
JST PH 3-pin comm cable 50 cm	REV-31-1418	4
JST PH 3-pin comm cable 100 cm	REV-31-1565	2
goBILDA encoder cable 600 mm (JST PH to XH)	3801-0709-0600	4 (in use on drive motors)
goBILDA JST VH to 3.5mm bullet motor adaptor 100 mm	3801-0613-0100	4 (in use)
AndyMark 12 in motor encoder cable	am-3926b-1200	3
AndyMark 12 in motor power wire	am-5198-1200	3
Zip ties 160 mm	REV-41-1161	100
Cable ties 4 in	am-1589	100

Gap: the 117 RPM goBILDA motor and the REV/AndyMark motors need their own power and encoder leads. The goBILDA motor ships with bullet leads; you have zero spare goBILDA bullet-to-JST VH adaptors and zero spare goBILDA encoder cables. Order 2 to 4 of each (3801-0613-0100 and 3801-0709-0600) when buying anything else from goBILDA.

6. Structure

goBILDA (in the built chassis)

Item	SKU	Qty
1120 U-Channel 10 hole, 264 mm	1120-0010-0264	2
1120 U-Channel 17 hole, 432 mm	1120-0017-0432	2
Quad Block Pattern Mount	1201-0043-0002	8
M4 thread-locking screws 8 mm / 10 mm	2822-0004-0008 / -0010	25 / 75 (mostly used)

REV 15 mm system (FTC Starter Kit V3.1)

Item	SKU	Qty
15mm Extrusion 420 mm	REV-41-1432	16
15mm Extrusion 225 mm	REV-41-1431	8
15mm Extrusion 120 mm	REV-41-1568	10
45x15 mm C Channel 408 mm	REV-41-1762	4
45x15 mm C Channel 248 mm	REV-41-1767	2
U Channel Endcap	REV-41-1687	4
Corrugated plastic sheet 450x300x4 mm	REV-41-1839	2
Plastic 90° bracket	REV-41-1305	32
Metal 90° bracket V2	REV-41-1480	16
Plastic 60° / 45° / 30° / 120° brackets	REV-41-1306/1307/1308/1311	8 / 8 / 8 / 6
Inside corner bracket	REV-41-1320	16
Lap corner bracket	REV-41-1321	24
Motion bracket / Rod end bracket	REV-41-1303 / -1304	8 / 8

Item	SKU	Qty
Gearbox motion bracket	REV-41-1315	4
Variable angle bracket	REV-41-1318	4
Plastic servo bracket / Metal bent servo bracket	REV-41-1319 / -1485	2 / 4
Bent Core Hex motor bracket	REV-41-1433	8
UltraPlanetary outside / bent mounting brackets	REV-41-1621 / -1623	4 / 4
Battery holder plate	REV-41-1166	2
15mm Hinge kit	REV-45-1259	1
M3 standoffs 30 / 40 / 45 mm	REV-41-1491/1492/1493	8 / 8 / 8
M3 hardware		500 × 8 mm, 100 × 16 mm, 100 × 20 mm, 50 × 35 mm screws; 500 nyloc nuts; 50 T-slot screws

AndyMark Robits (1/2 in grid, 10-32 hardware)

Item	SKU	Qty
0.5 in square tube: 2 / 4 / 6 / 8 / 10 / 12 / 14 / 16 in	am-5001-xxxx	4 / 4 / 4 / 4 / 6 / 4 / 2 / 2
1 in square tube 15.5 in	am-5002-1550	4
Beams 1×4 / 1×8 / 1×12 / 1×16, slotted	am-5011	4 each
Gussets: 45°, 135°, 2×3 L, 3×3 corner, 5×5 corner, 3×3 U	am-5010 / 5004 / 5005 / 5008	4 / 4 / 8 / 8 / 4 / 4
Plates 2×4 / 4×4 / 4×8	am-5006	4 / 4 / 2
Perforated polycarbonate 15.5 × 6 × 1/32 in	am-4964a	2
Motor mounts (80-100T, side 2-hole)	am-5017 / am-5012	3 / 1
Bushing and shaft carriers (end, side, double)	am-5013/5014/5015/5016	4 each
10-32 standoffs 1 / 2 / 4 in	am-1701/1702/1704	4 / 8 / 4
10-32 SHCS 1/2 to 2-1/4 in, nylock jam nuts, washers		~340 screws, 100 nuts, 100 washers

Item	SKU	Qty
Nylon and aluminum spacers		24 of each of 6 sizes

7. Motion components

REV 5mm hex (Starter Kit)

Item	SKU	Qty
Hex shaft 75 / 90 / 135 / 400 mm	REV-41-1347/1348/1349/1362	8 / 12 / 4 / 4
Plastic gears 15T / 30T / 45T / 60T / 72T / 90T / 125T	REV-41-133x	8 / 4 / 4 / 8 / 4 / 4 / 2
#25 sprockets 10T / 15T / 20T / 40T	REV-41-1338/1339/1340/1343	8 / 4 / 4 / 2
#25 roller chain 10 ft, 5 master links, chain tool	REV-41-1365 / -1366 / REV-42-1442	1 / 5 / 1
High strength hex hub	REV-41-1147	8
Bearing pillow block / hex pillow block	REV-41-1317 / -1316	32 / 4
Through-bore bearings short / long	REV-41-1326 / -1329	24 / 24
5mm hex bearing block	REV-41-1683	2
Shaft collars	REV-41-1327	40
Hex shaft spacers 15 / 3 / 1.5 mm	REV-41-1323/1324/1325	24 / 48 / 28
Tensioning bushings 39 mm	REV-41-1702	8
Surgical tubing 3 mm	REV-41-1163	1

AndyMark 3/8 in hex (Robits)

Item	SKU	Qty
3/8 hex shaft 3 / 4 / 6 / 8 / 10 in	am-5003	2 / 4 / 2 / 1 / 1
5 mm hex to 3/8 hex shaft adapter	am-4957-0100	4
20DP plastic gears 30T / 40T / 50T / 60T	am-5020	6 / 6 / 2 / 2
HTD 5mm 14T pulley halves	am-4960_half	16 (8 pulleys)

Item	SKU	Qty
HTD 5M belts 45T / 60T / 75T / 121T	am-5209	2 each
Motion bushings 3/8 hex	am-5021	24
#33 rubber bands	am-5024	50

goBILDA 8mm REX (in chassis)

Item	SKU	Qty
Shouldered 8mm REX hub-shaft, 20 mm offset	2110-0608-0002	4 (in use)
Clamping steel miter gear 24T	2320-4008-0024	8 (in use)
8mm ID spacers 9 mm, flanged bearings 6 mm / 8 mm ID, shims	1522-0010-0090, 1611-0514-0006/0008, 2807-0811-0500	4 / 4 / 4 / 12 (in use)

8. Wheels and rollers

Wheel	SKU	Qty	Bore	Use
GripForce Mecanum 104 mm, 40A rollers	3625-0202-0104	4	8mm REX	Drivetrain (installed)
90 mm Grip Wheel	REV-41-1267	4	5mm hex	Flywheel candidates
90 mm Traction Wheel	REV-41-1354	2	5mm hex	Flywheel candidates
90 mm Omni Wheel	REV-41-1190	2	5mm hex	Guide rollers
1 in Grip Wheel, soft	REV-41-2150	8	5mm hex	Feeder / indexer rollers
Flap Wheel, medium	REV-41-2702	8	5mm hex	Intake roller alternative
2 in Compliant Wheel, green	am-3571_green	4	3/8 hex	Intake or feeder
3 in Omni Wheel	am-4967a	2	3/8 hex	Guide rollers

Wheel	SKU	Qty	Bore	Use
3 in Stealth Wheel	am-4718_gray	2	3/8 hex	Flywheel candidates
TPU intake spoke wheels (3D printed)	`hardware/intake-spokes/`	as printed	8mm REX	Intake

9. Tools on hand

Wera 3 mm ball-end hex key, 1.5 mm and 2 mm Allen wrenches, 2 × 5.5 mm nut drivers, 5.5 mm combination wrench, 5/32 in and 2.5 mm ball-end hex drivers, 2.5 mm L-wrench, 3/8 in combination wrench, #25 chain tool, Robits ruler, two storage totes.

What this means for the first robot

- **Actuators are not the constraint.** 12 motors and 10 servos on hand, 8 + 8 allowed. Iteration 1 needs about 7 motors and 3 servos.
- **Three build systems that don't bolt together.** goBILDA (M4, 8mm REX), REV (M3, 5mm hex, 15 mm extrusion), and Robits (10-32, 3/8 hex, 1/2 in grid). Plan on 3D-printed adapter plates where a REV or Robits mechanism meets the goBILDA chassis. The 5 mm to 3/8 in hex adapters let REV motors drive Robits gears and belts.
- **Flywheel launcher is buildable from the box.** An HD Hex motor run bare (6000 RPM) or through a single 3:1 UltraPlanetary stage, driving 90 mm REV grip or traction wheels on 5 mm hex shaft, with the pillow blocks and bearings from the Starter Kit.
- **Intake is covered.** The printed TPU spokes on a REV 5 mm hex shaft driven by the Core Hex motor, all from the Starter Kit.
- **Vision is covered.** The C920 on a Control Hub USB port is exactly what R707/R708 allow.
- **Buy list before the launcher build:** goBILDA motor cables and encoder cables (see section 5), a second webcam mount or printed bracket, and whatever REX shaft length the intake needs (the chassis kit includes none spare). Everything else for iteration 1 is on the shelf.